For the Milestone 2 portion of the project, we were asked to conduct an evaluation of our end-to-end face verification pipeline. The primary objective of this phase was to transition from experimental scripts into a reproducible backbone capable of tracking experiments, managing configurations via YAML, and providing automated error analysis. We evaluated the system's ability to distinguish between pairs of images of the same or different individuals using the Labeled Faces in the Wild dataset.

The pipeline utilizes a configuration-driven architecture to ensure portability and reproducibility. We used a Cosine Similarity metric on raw pixel intensities, reshaped to 32 x 32 sized vectors. To ensure the statistical integrity of our results, we implemented a structured experimental loop consisting of five tracked runs. Each run was executed using a dedicated CLI command (via argparse), which automatically logged the results to a unique timestamped JSON record. This allowed us to iterate rapidly through different normalization strategies and data limits without manual record-keeping.

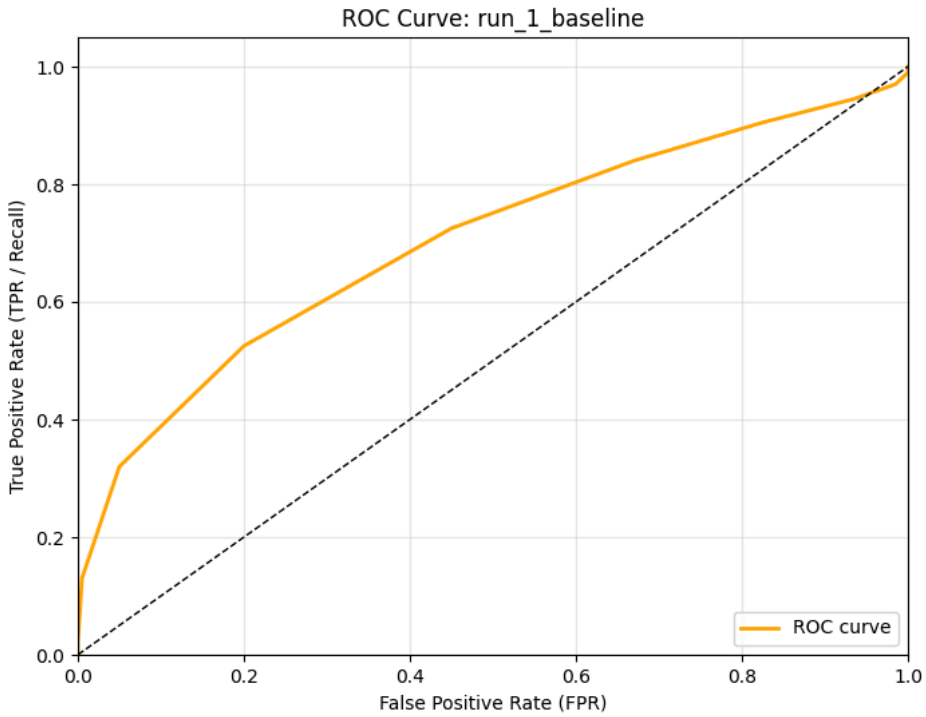
We conducted a sweep of five experiments to identify the impact of preprocessing and selection rules on model performance. Run 1 established a raw baseline with a Balanced Accuracy of 0.4950, confirming that unnormalized pixels perform at near-random chance. While Run 4 (Min-Max normalization) yielded identical results due to the scale-invariance of Cosine Similarity, Run 3 (Z-score normalization) provided our most significant boost, reaching a peak Balanced Accuracy of 0.5875. The threshold selection logic proved as critical as the data itself. We utilized a decision threshold of 0.2653 for our best run (#3) based on the Balanced Accuracy rule. This strategy was chosen to weigh the True Negatives and True Positives equally, which is essential for a biometric system where rejecting an imposter is just as critical as recognizing a valid user. Unlike the other runs, the F1-Score optimization used in Runs 2 and 5 consistently biased the model toward acceptance. In the extreme case of Run 5, the threshold collapsed to -1.0, causing the system to predict same for all of the pairs. While this yielded a mathematically higher F1-score (0.66), the 0% True Negative rate confirmed that the model had no discriminative power.

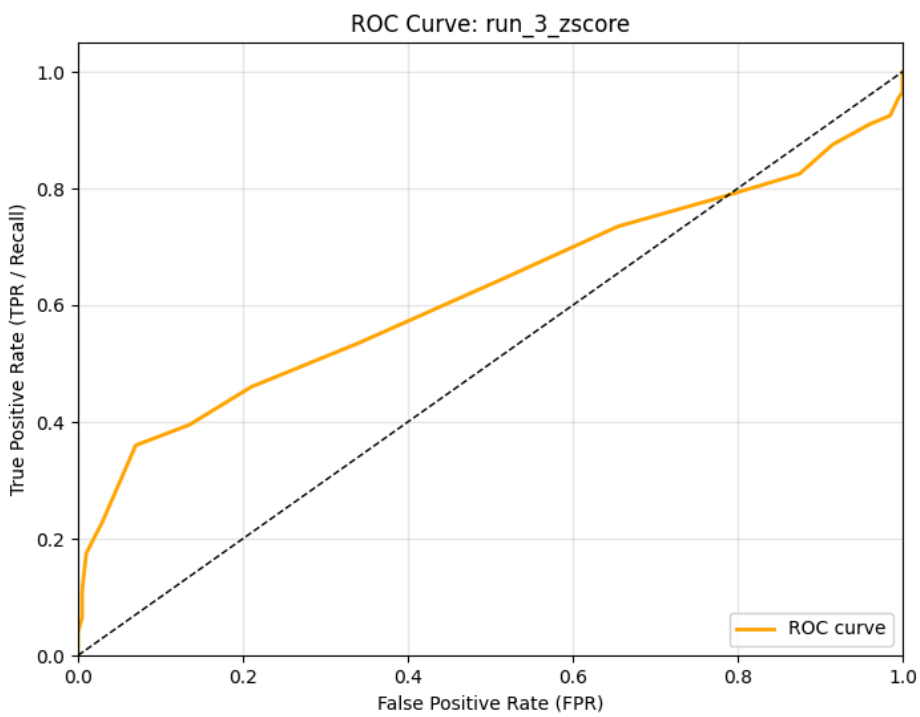
Run Summary

Run ID	**Normalization**	**Selection Rule**	**Balanced Accuracy**	**Key Observation**
Run 1	None	Balanced Acc	0.495	Near-random chance performance.

Run 2	None	F1-Score	0.5025	Improved Recall, but doubled False Positives.
Run 3	Z-Score	Balanced Acc	0.5875	Best separation of classes.
Run 4	Min-Max	Balanced Acc	0.495	Scaling didn't change the vector angles.
Run 5	Z-Score	F1-Score	0.4820	Threshold at -1.0, so the model accepted everyone.

Interestingly, while Run 3 provided the highest accuracy at a specific point, its ROC curve remained closer to the diagonal than the baseline. This suggests that while Z-score normalization allowed us to find a more effective individual decision threshold, it also increased the overall overlap between the "Same" and "Different" distributions. This likely occurs because centering raw pixels can inadvertently minimize the unique variances that distinguish individual faces, pushing the model toward the random-guess diagonal despite a "luckier" individual threshold.


ROC Curve: run_1_baseline
True Positive Rate (TPR / Recall)
False Positive Rate (FPR)
ROC curve
0.0
0.2
0.4
0.6
0.8
1.0


ROC Curve: run_3_zscore
True Positive Rate (TPR / Recall)
False Positive Rate (FPR)
ROC curve
0.0
0.2
0.4
0.6
0.8
1.0

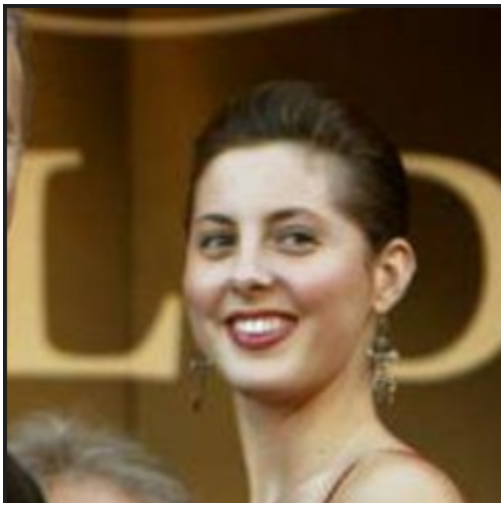
Run 3 (Z-score) Confusion Matrix

	Predicted: Different	**Predicted:** Same
Actual: Different	175 (TN)	25 (FP)
Actual: Same	140 (FN)	60 (TP)

Error Analysis

By examining the automated error slices generated by the pipeline, we identified two critical weaknesses in the current pixel-based approach. In our False Positive analysis, we observed that individuals like Aaron Guiel were mistakenly matched with others who shared similar skin tones or open-mouth poses, such as Emily Rossum and Howard Smith. This suggests the model is currently acting as a color detector/analyzer rather than a face recognizer; because raw pixel values for specific lighting or complexions are mathematically similar, the similarity score stays high regardless of facial geometry.

Our False Negative analysis further highlighted the model's sensitivity to environmental noise. We observed that individuals like Aaron Sorkin and Ai Sugiyama, were rejected when lighting conditions or backgrounds differed in images of them. In the case of Sugiyama, the presence of a tennis visor introduced shadows that the current model could not distinguish from actual facial features. Because raw pixels give equal mathematical weight to the background and the face, any shift in lighting effectively relocates the image to a different region, causing the system to fail.



The most important lesson from this iteration loop is that infrastructure precedes intelligence. By building a reproducible pipeline, we proved that Z-score normalization improves precision by

nearly 20% by centering the data. However, the error analysis confirms that raw pixel intensities are fundamentally insufficient for true face verification, as they are too easily fooled by lighting and skin-tone noise.